EXPERIMENT NO:09

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import r2_score

# Dataset
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([5, 8, 15, 25, 40, 60, 85, 115, 150, 190])

# Linear Regression
linear_model = LinearRegression()
linear_model.fit(X, y)
linear_pred = linear_model.predict(X)

# Polynomial Regression
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)
poly_pred = poly_model.predict(X_poly)

# Display results
```

```
print("Linear Regression R2 Score:",
      round(r2_score(y, linear_pred), 4))

print("Polynomial Regression R2 Score:",
      round(r2_score(y, poly_pred), 4))

# Smooth curve for plotting
X_curve = np.linspace(1, 10, 200).reshape(-1, 1)
linear_curve = linear_model.predict(X_curve)
poly_curve = poly_model.predict(poly.transform(X_curve))

# Graph 1: Compare regression models
plt.figure(figsize=(9, 5))
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X_curve, linear_curve, color='red',
         label='Linear Regression')
plt.plot(X_curve, poly_curve, color='green',
         label='Polynomial Regression')
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.grid(True)
plt.show()

# Graph 2: Compare R2 scores
```

```
scores = [
    r2_score(y, linear_pred),
    r2_score(y, poly_pred)
]

plt.figure(figsize=(7, 5))
plt.bar(['Linear', 'Polynomial'], scores)
plt.ylabel("R2 Score")
plt.title("Regression Model Performance")
plt.ylim(0, 1.05)

for i, score in enumerate(scores):
    plt.text(i, score + 0.02, f"{score:.3f}", ha='center')

plt.show()
```

OUTPUT:

```
Linear Regression R2 Score: 0.92
Polynomial Regression R2 Score: 0.9999
```



